Bachelor Degree Project:

Superconductivity in 2D Multi-components Materials

Time-reversal symmetry breaking in 2D multi-components material using the Monte Carlo simulation

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Year: 2023

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Abstract

The purpose of this project is to determine the statistical properties of superconductors that break multiple symmetries in ground-state 2D multi-component superconductors depending on their superconductive state and above states. This is meaning the study of phase transition, determining the physical form of order parameters which turns out to explain whether time-reversal symmetry and phase stiffness are affected. These characteristics are also related to frustration in the phase transition which are determined by the inter-band coupling terms.

Hence, this project is about writing a Python code related to statistical mechanics in a multi-component XY model using the Monte Carlo method with a Metropolis algorithm. This program will plot the corresponding order parameter related to time-reversal symmetry. Indeed, the temperature will be the parameter allowing us to observe time-reversal symmetry breaking and normal state, which explains one of the two characteristics of superconductivity, the second being phase stiffness.

Note that this work is taking a precise scenario: A 2D multi-component with three identical bands with a frustrated Josephson coupling in the London limit and without gauge field.

Contents

1 Introduction: Theory of superconductivity

1.1 What makes a multi-component superconductor

A current flowing through a conductive material in a normal state is subject to a resistance determined by Ohm's law: $R = \frac{U}{I}$, R being the resistance, U the current, and I the intensity. However, this was until 1911 when superconductivity was discovered at the Leiden Laboratory [1]. Indeed, superconductivity is a state where there is no resistance in the material crossed by the current.

The electrons giving rise to the current have been long time thought of as a particles bouncing back and forth between atoms. But Quantum mechanics is saying that those electrons are acting as waves. Hence, Felix Bloch gave an elegant answer in 1930 to the behavior of the electrons as a wave between the atoms of a metal. Bloch found out that, as the atoms are placed periodically in the material (in lattices), an electron doesn't have any choice but to adopt a periodic wave to be coherent with atoms. Hence, the electron can have a sophisticated wave function but have to be repeated periodically in agreement with the atoms. This wave can cross the entire metal and can move like the atoms weren't there. It's only when an atom is placed where it shouldn't be, that the wave is bothered. That is, the resistance of the current in metal is due to the imperfections of the lattice: atoms that are found to be where they shouldn't. And a perfectly crystalline metal would engender a perfectly conductive material without resistance. Nevertheless, this will not be sufficient to make the material 'super'-conductive.

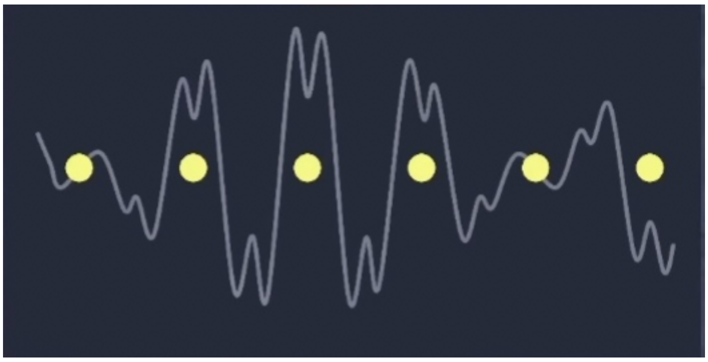


Figure 1: A periodic electron's wave function as a plane wave in a lattice of atoms described by Bloch in 1930, taken from [2]

However, this is applying to a single electron. We will see that a superconductive state is related to a wave function coherent for several electrons.

But, what separates a regular material from a superconductive material? First of all, the superconducting state is more specific when the order parameter is building a coherent wave function throughout the material. The order parameter is defined by when electrons form Cooper pairs (pairs of electrons that form a shared state) and condense into a macroscopic wave function.

Omitting the electromagnetic potential, we can construct an order parameter of the form:

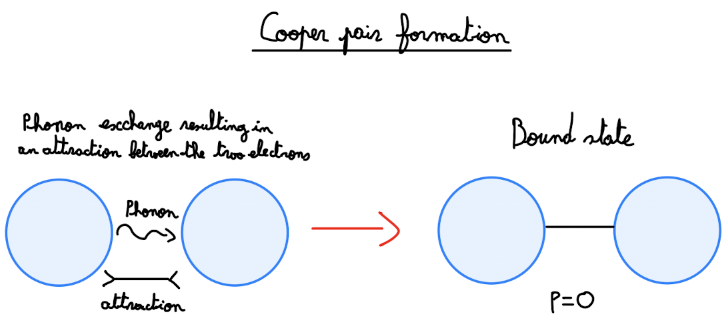
$$O_s = \int \frac{\Psi}{A} dx$$

Ψ being the wave-function and A being the Area over which the wave-function is spread.

The superconducting order parameter is a complex quantity: $\Delta(p)e^{-i\phi(x)}$. Where $\Delta(p)$ describes the magnitude of the superconducting gap (roughly the energy required to break a Cooper pair), which may have some momentum p-dependence around the Fermi surface. The other quantity in that expression is the phase $\phi(x)$.
Superconductivity is marked by phase coherence like a laser or a Bose-Einstein condensate, but with Cooper pairing. Hence, a superconductor will choose an arbitrary phase (possibly with some non-arbitrary x-dependence)[3].

The electrons pair up and move throughout the component without resistance, which is resulting in a state with zero electrical resistance. We will see that this state is maintained as long as the temperature of the superconductor is below a certain critical temperature.

The process of Cooper pair formation involves the exchange of a phonon resulting in an attraction between the electrons, which is forming a bound state with zero total momentum.
These pairs condense so that they can be described by the same wave function as they are in the same quantum state(Cooper pairs are bosons in their ground state) , this is called a condensate.



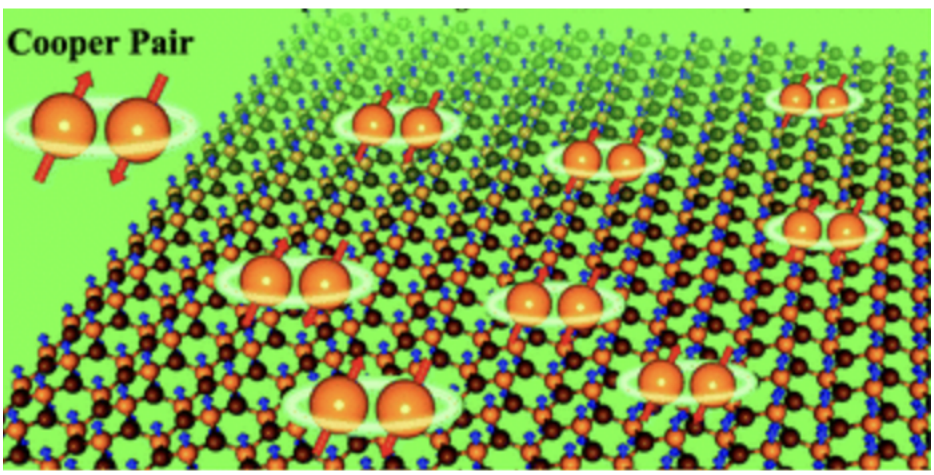


Figure 2: Superconducting 2D Phosphorus Carbide, taken from [4]

We model the system with no gauge field, which corresponds to be type II limit. In this work, we are inspired by 2D superconductivity rising in moiré materials, like twisted bilayer graphene. Moiré materials are the superposition of two thin lattices stacked at small relative twist angles. This is resulting in a picture of interference patterns and thus fringes. Indeed, when twisted from one to another, 2D graphene is showing isolating and metallic characteristics. But at $1.1°$, electrons are describing a macroscopic wave function that can be influenced by a magnetic field. That is, two 2D graphene stacks are showing a superconductive behavior when twisted from $1.1°$ approximately. The phenomenon is due to a spatial interference effect between the two lattices causing a phase alignment at a certain angle and low temperature. This field of study is called "twistronics" and is efficient at low temperatures.

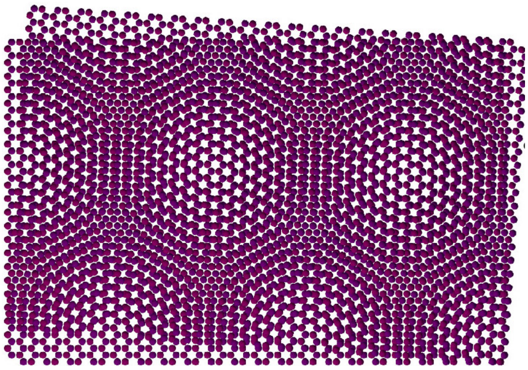


Figure 3: Twisting of two 2D graphene from one to another, taken from [5]

Until now we assumed that the order parameter and hence, the superconductive state are homogeneous along the material, however, this is not the case. The wave function is described by the coherence length (often written ϵ), the scale over which that superconducting order parameter varies depending on the electron-phonon interaction(Cooper pairs), the density of states at the Fermi level, the magnetic field(which will not be treated in this work). The coherence length is the characteristic length scale of fluctuations in the superconductor order parameter. Quantum fluctuations are the temporary random change in the amount of energy in a point in space, following Heisenberg's principle: $\Delta E \Delta t = \frac{\hbar}{2}$, E being the energy, t the time and $\hbar$ Planck's constant [6]. Here, we go beyond mean-field theory (describing the stochastic model by a simpler model averaging over degrees of freedom) to describe non-local fluctuations in the wave function.

This heterogeneity is also related to the phase, the relative alignment of the superconducting order parameter in different parts of the material. That phase will be at the center of this study, to determine the type of symmetries that the material is showing in different scenarios.

1.2 Josephson coupling in the London limit

But before introducing the symmetries, a focus on the setup built for this project surely needs to be made.

Our 2D multi-component superconductor exhibits a Josephson Coupling, that is, two of these components can be coupled together. The nature of this coupling is quantum tunneling generating a weak link between those two. This quantum tunneling is the action of the particle passing through the potential even if the binding energy is higher than the potential of this particle (electrons in our case)[7]. Precisely, the Josephson will remain stationary in our scenario, call that the dc Josephson effect.

In superconductivity, when a Josephson junction is crossed by a current, the electrons of the two superconductors are merging into one single quantum body through tunneling[1], meaning that the electron from both sides is forming a coherent single wave function.

In the superconductor, the free energy can be described by the Ginzburg-Landau free energy:

$$f = \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{2m_{cp}}|(-i\hbar\nabla - 2e\vec{A})\Psi|^2 + \frac{\mu_0 B^2}{2}$$

With f the free energy, B the magnetic field, μ_0 the vacuum magnetic permeability, e the charge of an electron, m_{cp} the mass of Cooper pairs, $\vec{A}$ the potential vector when a gauge field is present and $|\Psi|^2$ is the concentration of electrons in the superconducting state, that is to say, the order parameter that can be written $O_s = |\Psi|^2$.

We work in 2D where screening can be neglected, meaning that coupling to gauge field can be ignored. This gives free energy of the form:

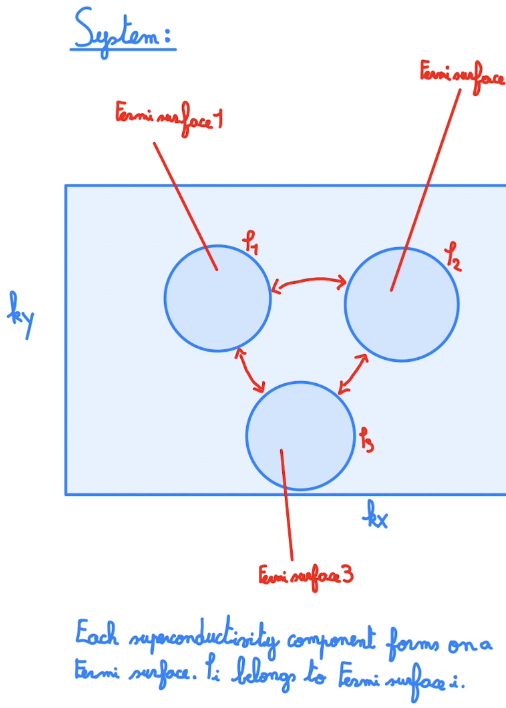
$$f = \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{2m_{cp}}|\nabla\Psi|^2$$

Where we take $\hbar = 1$. We also assume constant density, corresponding to the London limit of superconductivity.

In a system with multiple Fermi surfaces several superconducting components may arise (see image).
We model this scenario with a free energy of the form,

$$f = \alpha_i|\Psi_i|^2 + \frac{\beta_i}{2}|\Psi_i|^4 + \frac{1}{2m_{cp}}|\nabla\Psi_i|^2 + \epsilon_{ij}\Psi_i\Psi_j^* + constant$$

Where the last term $\epsilon_{ij}\Psi_i\Psi_j^*$ is due to inter-band tunneling and i, j are corresponding to the different Fermi surfaces.
When $\epsilon_{ij} > 0$, the inter-band term is minimal when the phase difference between the components is π. For three components, this condition cannot be simultaneously satisfied for all components, resulting in frustration. Frustration refers to atoms that tend to stick to non-trivial positions, on a regular crystal lattice, conflicting inter-atomic forces[8]. In this scenario, the ground state may break time-reversal symmetry.



We use this description with periodic boundary conditions, on a 2D square lattice. Also, we will be able to witness when this Josephson coupling will impose spontaneously broken symmetry. That is, when the system is reaching the critical temperature T_c separating normal and superconductive states. At this temperature, T_c, two symmetries defining superconductivity will also break spontaneously from the normal state. Thus, let's have first and foremost a recall of the characteristic symmetries of the superconductor.

1.3 Symmetries

Key characteristics of the superconductive state are its stiffness and time-reversal symmetry.
Globally, symmetries are representing attributes and intrinsic characteristics that are remaining unchanged in space-time after one or several transformations. In Quantum mechanics, we assume that there are no causal laws justifying physical processes, but the behavior of quantum bodies is described only by probabilities restricted and constrained by symmetries[9].

Introducing two superconductivity properties: time-reversal symmetry and phase stiffness, characterizing the system built statistically. In this work, only the time-reversal symmetry is being simulated numerically, but the phase stiffness (with $U(1)$ symmetry) is presented. Hence, we will not fully characterize superconductivity.

1.3.1 Time-reversal symmetry

The second law of thermodynamics is stating that the entropy can only increase in a closed system when time is going forward, that is to say, the macroscopic scale is not showing time-reversal symmetry, it is then asymmetric (even though for equilibrium states, this law predicts the symmetry to hold)[10]. Time-reversal symmetry means that the laws of physics are the same whether you look at a system going forward or backward in time[11]. However, in quantum micro-states, this law is said to be broken.

In quantum mechanics, the time-reversal attribute is associated with the operator T, which is defined by the equation:

$$T\Psi(x,t) = \Psi^*(x,-t)$$

This algebraic operator is "reversing" a few intrinsic properties of a quantum body such as the momenta and the total angular momentum:
$TpT^{-1} = -p$ (=anti-commute)
$TJT^{-1} = -J$
But is however not acting on the position: $TxT^{-1} = x$ (=commute)

More generally, the definition of T is an anti-unitary operator, meaning that the operator can be written as $T = UK$, where U is a unitary operator and K is the complex conjugate operator ($KcK = c^*$, $c \in \mathbb{C}$).

T is also characterized by Kramer's theorem stating that $T^2 = \pm 1$, by showing $T^2 = UKUK = UU^* = U(U^\dagger)^{-1} = \Phi$, where Φ is a diagonal matrix of the phases. The above calculation is causing $U = U^\dagger\Phi$ and $U^\dagger = U\Phi$, then we conclude $U = \Phi U \Phi$ which is meaning that Φ is either equal one or minus one.

Hence, in certain classes of frustrated multi-component superconductors, time-reversal symmetry is spontaneously broken. This is meaning that the corresponding order parameter is modeled under time-reversal symmetry. Broken time-reversal symmetry means gauge symmetry is not invariant under time-reversal. In the context of GL-theory, broken time-reversal symmetry is manifested in a non-vanishing order parameter S, satisfying $TS = -S$ (that is $\langle|S|\rangle \nrightarrow 0$ in macroscopic limit).

The purpose of this work is to show and explain why this symmetry is broken. Nevertheless, it is important to introduce both phase stiffness and $U(1)$ symmetry as well.

1.3.2 Phase Stiffness(2D), $U(1)$ symmetry(3D)

When modeling a superconductor without a gauge field, superconductivity is associated with a finite phase stiffness, meaning that phase gradient costs energy. In 2D, continuous symmetry is not broken due to the Mermin-Wagner theorem and superconductivity has to be characterized by phase stiffness. The phase stiffness is identified by the energy required to change the phase order, resulting in the distortion of the lattice. In 3D, this phase stiffness results in spontaneously broken $U(1)$ symmetry. The term "U(1)" comes from the mathematical group that represents this symmetry, which is a unitary group with one generator.

As said before, a superconductor can be seen as Cooper pairs that make up the condensate and a single macroscopic entity that behaves like a single wave function with the number of particles in the condensate being conserved. These Cooper pairs are affecting the distribution of the charges. The $U(1)$ symmetry is treating of a gauge symmetry conserving the veracity of the laws of physics, under some transformations to the fields and particles of the system. Spontaneously broken $U(1)$ symmetry implies a finite phase stiffness. Hence this symmetry is related to the phase and is affected by the electric charge of the particles.

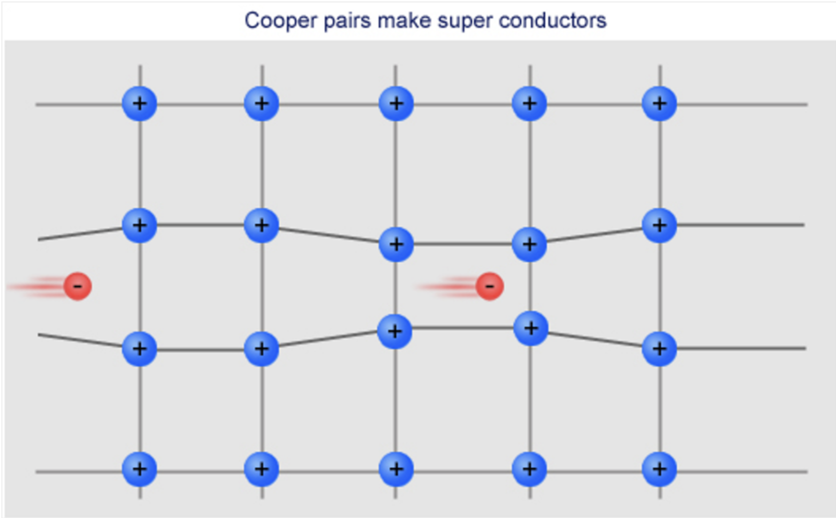


Figure 4: Illustration of how deformation of the ion lattice mediates attraction between electrons, leading to the formation of Cooper pairs and thus superconductivity, taken from [12]

The phase stiffness is related to the London equation, throughout the London penetration depth. The penetration depth is a measure of how deep a magnetic field can penetrate a superconductor before being screened by superconductor currents. The London penetration depth is inversely proportional to the square root of the stiffness of the superconductor $\lambda \propto \frac{1}{\rho_s^{\frac{1}{2}}}$, where λ is the London penetration depth and ρ_s is the superconducting stiffness. That is meaning that a component with a small London penetration depth is related to a high stiffness and causes an effective screening of magnetic fields. Hence, the stiffness and the London penetration depth of a superconductor will predict its behavior when it encounters a magnetic field.

Superconductors showing high stiffness and small London penetration depth are often used in the presence of a magnetic field to build useful instruments such as magnetic resonance imaging (MRI) machines, and particle accelerators.

The superconducting stiffness ρ_s can be measured by the London equation:

$$\vec{j} = -\rho_s \vec{A}$$

$\vec{j}$ being the current density and $\vec{A}$ the potential vector. A value of the stiffness can be measured by applying a rotor free $\vec{A}$ and by measuring the current density j using the magnetic moment of a ring[13].

However, this measurement will not be performed in this work.

These two symmetries are related to the macroscopic phase of the system. Microscopically, a superconductor is typically nucleated from a Fermi liquid state. At the onset, the Fermi surface becomes unstable as Cooper pairs form. The ground state of a superconductor is no more a Fermi gas/liquid, but a Bose gas/liquid (forming a Bose-Einstein condensate), the Fermi surface is no more conserved[14] and the system gapes. This instability of the Fermi surface is the hallmark of superconductivity as described by Bardeen, Cooper, and Schrieffer (BCS), in the BCS-theory. Hence the spontaneously broken time-reversal symmetry and phase stiffness are also associated with the instability of the Fermi surface.

To imagine such a physical behavior scenario we numerically approach experimental values using statistical methods. Given a maximum number of simulations, we collect the data, to potentially extract meaningful results. The steps and the details used for this numerical approach are treated in the below section.

In this work, we examine the statistical properties of a superconductor that also break spontaneously time-reversal symmetry in the ground state. Specifically, we focus on how time-reversal symmetry is restored by thermal fluctuations.

2 Methods: Statistical Mechanics, Monte Carlo Simulation, Metropolis Algorithm

2.1 Statistical Mechanics

We need to introduce Boltzmann statistics to answer this problem at finite temperatures, and then characterize a probability to obtain an order parameter across the system. We are using the Boltzmann factor to define how the system adopts a resulting macro-state proportional to the number of micro-states. Then this factor is quantitatively giving the probability to find a certain macro-state.

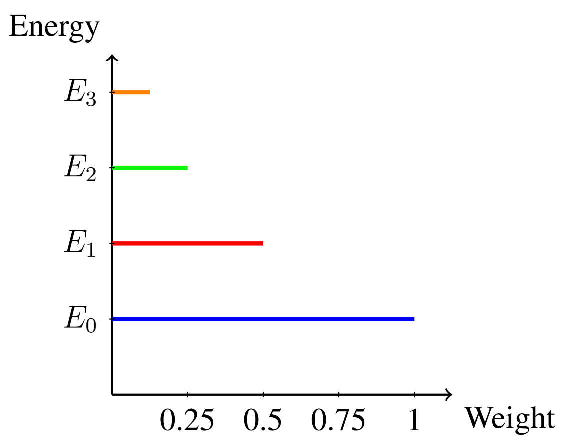
Hence, the Boltzmann factor being the probability of finding the particle in one energy state under the macroscopic scale, we have statistically,

$$P(E_i) = \sum_i e^{-\beta H_i}$$

β being one over the k Boltzmann constant and the temperature T: $\beta = \frac{1}{kT}$. In the numerical work, k is set to be one to simplify calculations. And i is here the energy level.

The partition function is counting how many states are accessible weighting each one in proportion to its probability[15]. At zero temperature the system will be in its ground state. And when the temperature increases, this probability is staying lower than one. The probability is taking a zero value when the temperature is increasing just enough from its zero value. That is meaning that close to zero, we have no chance to find the energy state given. As well, you can note that when the temperature is precisely equal to zero, the probability is one (even though we cannot divide by 0 in the definition of β). As T is increasing we have a better chance to find this macroscopic energy, but will remain lower than when $T = 0$.

By analogy, the terms of the sum defined by this partition function are getting smaller as the Hamiltonian is increasing.
This probability could be represented over a diagram, for instance, we could have:



The level of energy of every "local" electron is related to the Hamiltonian and its Boltzmann factor.

The Hamiltonian contains contributions from local inter-band terms and gradient terms.

Discrediting the free energy equation in the London limit (from section 1.2), we obtain:

$$H = - \sum_{A,<i,j>} \cos(\varphi_{A,i} - \varphi_{A,j}) + \eta \sum_{i,A \neq B} \cos(\varphi_{A,i} - \varphi_{B,i})$$

The first term represents kinetic energy due to super-currents. The second term represents Josephson's inter-band tunneling and gives rise to frustration concerning phase difference. For a single phase A of a site i, the different phases are named B, and neighboring sites are called j. η is the amplitude for inter-band tunneling, a parameter related to the component and determining its superconductivity. It is set to be one in this work as we are not studying a precise multi-component.

Quantum mechanics and thermal statistics characterize the intrinsic properties of every single site to construct an accurate and physically correct lattice. These definitions are deciding with probability conditions the preferred macroscopic system. Hence, a probability-related simulation is also used to characterize numerically the condensed matter properties of our system.

2.2 Monte Carlo Simulations

From the rolling dice game of the casino of Monaco, the Monte Carlo method is a mathematical and statistical tool used to predict the statistical properties of a system. This is based on stochastic sampling to obtain statistical properties. This probabilistic model is including uncertainty and random in its prevision to simulate a possible final result.

The method is based on ergodicity, that is to say, the statistical behavior of a point in a closed environment, and that same point will end up having used every location of this ergodic system[16]. The Monte Carlo simulation is repeating a lot of times a scenario to keep the probable ones in the output. Ergodicity means that a system can in theory visit every point of configuration space.

We use the method to establish an average of accepted and rejected values from a random sampling of an initial and a proposed second state of a lattice site. For instance, we take random initial phases for one lattice site and then we propose a second random state for this site. Thus, under some conditions, the second set of phases is accepted or else we keep the initial parameters. These conditions are given by the Metropolis algorithm which will be detailed next.

This probability method is very important to take into count different factors such as local inter-band and gradient terms to keep the most probable phases in order. The simulation predicts how are spread the influence of each site on its neighbor over the system. As described in 2.1, the average order parameter is mainly directed by the temperature as the Boltzmann factor is depending on it.

The Monte Carlo method is taking the following form:

(1)-Input values:
Values for phases are first randomly generated as these can take any values possible. However, this method could take initial values given by a law of probability, but this is not the case here.

(2)-New example sample of values:
New values are again randomly proposed but are in general following a probability distribution.

(3)-Mathematical model:
To accept the proposal according to a statistical weight, we link the first and second generated values. We set up a rate of the input values and the new example values sample. The new values are accepted if the rate is following probability distribution which was determined upstream, and if not, the input values are kept in the overcome data.
Note that the mathematical model is generating a rate in our case but could generally be financial, economics, or mathematical equations[16].

(4)-Output the accepted values:
Using the mathematical model to obtain an accurate estimate of the observable, a sufficient number of measurements must be collected, whether these are from the initial values or the new example, the more accepted values we collect, the more the mathematical model will reflect on the final result.

(5)-Analysis of the data given:
Many useful statistical tools can be used to calculate parameters, such as standard deviation or median. An average of the accepted values can be made, as well as a plot to conclude. In this work, a plot of the average order of the phases for every single site is generated.

Indeed, this method is including probabilities and statistics in its prediction that is leading to uncertainty, meaning that we have different results every time we run the simulation. We are using probability densities to predict a possible appearance of a value. Those statistical functions are defined by a range of values contained between limits.

The Monte Carlo method can take several probability laws:
-Normal distribution: The simple symmetrical Gaussian curb, in which the standard deviation and median can be set up.
-Uniform distribution: A statistical representation of a random probability with an equal chance.
-Triangle distribution: Using the highest, lowest, and most probable values to represent random variables.

We are using in our case a uniform distribution, all the values have a chance between 0 and 1 to appear, as we cannot predict whether some phases are prioritized or not in a site of the lattice. We can only predict the spreading of prioritized values in neighboring sites and all over the component.

This method is asking a lot of data treating to show a concrete result. As well, the output values and the distribution probability are the main concern to get an accurate analysis. The simulation output several different results, using their probabilities to appear from a large sample of determined values, resulting in a much clearer prediction.

2.3 Metropolis Algorithm

In our method used, an argument for accepting or rejecting the new candidate sample is missing. The Metropolis-Hasting algorithm is taking into count the probability law chosen, the initial values, and the new sample generated. And then, implement a condition and decide which value is kept in the resulting sample.

The Metropolis-Hasting algorithm is given by the following[17]:

(1)-Generating/ Selecting an initial value, basically the input values of the Monte-Carlo are taken. This value will be called θ_0

(2)-The number of new samples studied being n (that is, the number of simulations steps), we repeat n times the following:

> Step 1: We draw a candidate, meaning that we generate a value for the second sample. This new value is called θ^*. θ^* is drew from a proposal probability distribution chosen: $p(\theta^*|\theta_n)$.
>
> Step 2: We set up a rate: $r = \frac{p(\theta^*)/p(\theta^*|\theta_n)}{p(\theta_n)/p(\theta_i|\theta^*)} = \frac{p(\theta^*)p(\theta_n|\theta^*)}{p(\theta_n)p(\theta^*|\theta_n)}$
>
> Step 3: Then the final condition to decide whether or not we keep the new value is:
>
> If $r > 1$, we keep the new value θ^*.
>
> If $1 > r > 0$, we have a probability r to also keep the new value θ^*.
> And then we have $1 - r$ probability of rejecting the new value and only the corresponding initial value θ_0 is kept.
>
> The value kept is the new initial value θ_n and is compared with a new proposed one in the next step, with n the corresponding simulation step number.

The meaning of this algorithm is that if the new values generated are more probable than the initial one, that is if $r > 1$ then we only take it to count the second values θ^*. If not, we have a probability to reject or not θ^* depending on the density probability chosen.

If the probability density chosen to characterize our new value is symmetrical, then we have the Hasting term equal to 1, $\frac{p(\theta_n|\theta^*)}{p(\theta^*|\theta_n)} = 1$. The probability density evaluated at the candidate given the initial value will be the same as the probability density evaluated at the initial value given the candidate and then will end up canceling each other in the ratio. Thus, we can highly simplify the calculus and our ratio will read $r = \frac{p(\theta^*)}{p(\theta_n)}$.

The Metropolis Monte Carlo method aims to generate a trajectory in phase space which samples from a statistical ensemble. The mathematical problem is then solved by a stochastic sampling experiment. The generation of random numbers is followed by arithmetic and logical operations[18].

The mathematical and theoretical parts of this numerical work have now been explained. In the next section, we treat more concretely the problem and how the simulation has been written. Moreover, the data acquired will be exposed and its accuracy will be discussed.

3 Results: The program

3.1 The code

Now that the context and the purpose of this project are setups, we can move on to deeper details on the program written to simulate the physical behavior of superconductors materials.

First and foremost, the code is randomly generating an initial lattice, that is to say, for a 5×5 lattice in our case, it is creating 3 phases for every site. Now having our 75 phases attributed to every single site, an empty list is settled, which will take every average order of the phases for all the sites given the temperature.

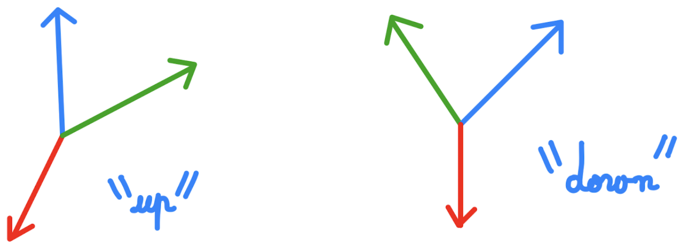
Next started with the simulation: the Monte Carlo statistical method. For a defined number of simulations at a given temperature, we generate/propose a new site with random phases. Hence, the program is taking into count a function called "prob" of the Hamiltonian H, returning $prob(H) = e^{-\frac{H}{T}}$. Referring to the Boltzmann factor given in the statistical mechanics section 2.1, note that we take the Boltzmann constant equal to one as we are just interested in the varying aspect of the function.

Staying in the simulation, we compute, for every single site, the energy corresponding to the initial and the proposed value. This is numerically the moment where the different phases of the site studied and the neighboring site's phases are acting. Hence, energies of the initial and proposed values are linked to the function "prob" given before, through the probability ratio $r = \frac{prob(H2)}{prob(H1)}$, if we call the energy from the initial value $H1$ and the proposed energy $H2$. This ratio refers to the rate used in the Metropolis algorithm.

Indeed, the Metropolis condition is yet to be set.
The ratio r is then compared to a random number x chosen between 0 and 1, which is the corresponding probability law chosen in 2.2, the uniform distribution. Thus, the value generated in the simulation is accepted or rejected under the Metropolis condition, but a final value is in any case kept. The new value is taken if $r > x$ (and if $r > 1$) and the initial one is taken otherwise.

Now we construct an order parameter to asses time-reversal symmetry breaking, by associating a value depending on the order of the phases. We consider that the site is "up" when the three phases are following a certain order, and is characterized by "down" if that order is not followed. More specifically, if we call the three phases of the site studied $p1$, $p2$, and $p3$ the program is returning "up" depending on if it is validating one of the three conditions:
-$p1 > p2 > p3$
-$p3 > p1 > p2$
-$p2 > p3 > p1$
Otherwise, the overall phase is considered as "down".

Example of phase order "up" and "down"



Finally, after keeping every value "up" and "down" in a list, under the form 1 and -1, the code is returning an average from the amount of simulation made off the phases kept. This average phase order is defined by:

$$S = \sum_i \frac{S_i}{N}$$

Where i is corresponding to the site concerned.

Also, every single average of the 25 order phases from the sites of the lattice is put in common and another average is made off the 25 values. Thus, a single final average called S is returned. And, at the end of the day, the program can plot a graph of $S(T)$, the average order phase of the whole lattice, depending on the temperature.

We used as well a method to make the program a bit faster: we only start collecting data after a defined number of simulations. That is, we slowly bring the system to its preferential configuration and only then do we collect the data for the final average.
Also, when collecting the data, we reuse the values acquired from the last steps to compare with the new one, so that it slowly influences the average and we can see the tendency of the system going to a favorite configuration.

Plotting $|S(T)|$ is allowing us to observe the extension of the order parameter related to time-reversal symmetry throughout the material. As presented in **??**, we are expecting a probability equal to 1 when the temperature is zero: that is causing the Metropolis algorithm to output the same phase order for the whole lattice. Hence, this is resulting in the spontaneously time-reversal symmetry breaking of the multi-component.

Theoretically, neighboring sites showing the same phase order are said to be correlated through Cooper pairs and share a common wave function. This is meaning that at $T = 0$, we expect a wave function to take over the entire 2D lattice, that is to say, a spontaneous time-reversal symmetry-breaking state extended to the entire material. Nevertheless, phase stiffness stays an aspect that needs to be studied to characterize superconductivity.

This program is also able to plot an XY model representing the phase order for every single site at the corresponding location for a given temperature.

The resulting data from this program is presented and studied in the following section.

3.2 Results

We run the program several times with different parameters to get more precise results. These parameters are the number of simulations chosen for every defined temperature, and the step of the temperature. Then the program draws slowly the graph at every temperature step. The more steps we have the more precise result we have. However, the running time can be tremendously long. For instance, the program took nearly one day to output the plot for five thousand simulations every 0.05 temperature step.

We calculate the order parameter S related to time-reversal symmetry. This order does not directly measure superconductivity. That would require also phase stiffness, which is not calculated here. Thus, we can report only on time-reversal symmetry breaking.

Here we have the graphs acquired when running these simulations:

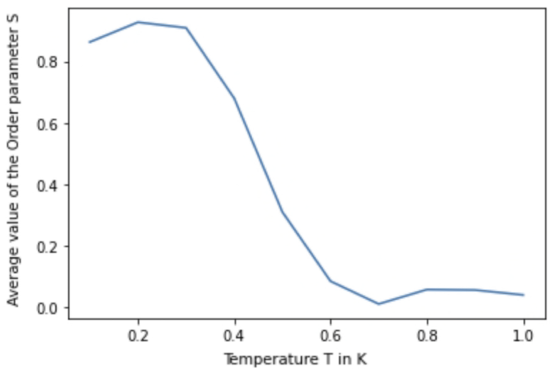


Figure 5: Graph of the average order parameter $\langle|S|\rangle$ related to time-reversal symmetry in the function of the temperature, for 1000 simulations every 0.1 temperature steps

In this figure 5, we could explain the dip at low temperatures by a system not fully reaching its equilibrium or statistical noise.

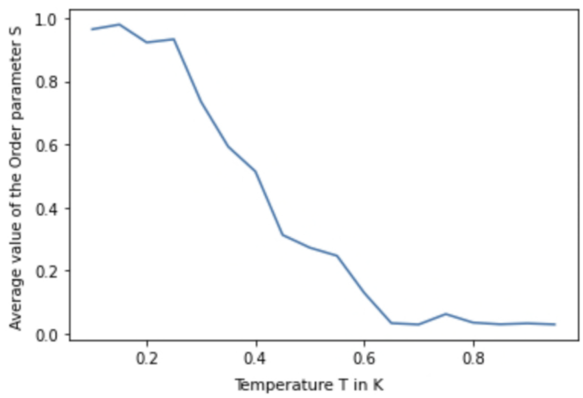


Figure 6: Graph of the average order parameter $\langle|S|\rangle$ related to time-reversal symmetry in the function of the temperature, for 5000 simulations every 0.05 temperature steps

If time-reversal symmetry is not broken, $\langle |S| \rangle$ is equal to 0 in the macroscopic limit. Reciprocally, if time-reversal symmetry is broken, $\langle |S| \rangle$ is different from zero in the macroscopic limit. Likewise, we could evaluate $T_C = 0.7K$ to separate the two states, but this value is not accurate since it would have needed to run the program for different lattices size (also with periodic boundary conditions), for instance, 4x4, 6x6, 8x8... To determine T_c from finite size systems, one has to do finite size scaling and simulate systems of varying size, but this is not attempted here. Nonetheless, we do see that the order parameter S decreases with temperature, indicating a phase transition.

We also see that longer simulations would be needed to see a smooth graph, implying that there are substantial statistical errors. This can be attributed to the fact that this was done in Python, as the running time of the program is not efficient for this type of numerical simulation.

This table is showing the precise values when calculating the average order parameter S for temperature going from $0K$ to $1K$ (Figure 4):

Temperature T in K	Average value of the Order Parameter S
0.1	0.9642
0.15	0.9792
0.2	0.9226
0.25	0.9326
0.3	0.7348
0.35	0.5924
0.4	0.5139
0.45	0.3128
0.5	0.2726
0.55	0.2469
0.6	0.1296
0.65	0.0335
0.7	0.0292
0.75	0.0623
0.8	0.0353
0.85	0.0299
0.9	0.0329
0.95	0.0295

Note that the values have been cut to the fourth decimal for easier reading.

We also modify the code to plot an XY-model, which is a representation of the phase order described in 3.1: a blue up arrow is defining an order of the three phases for a lattice site, while the red down arrow is the other order.

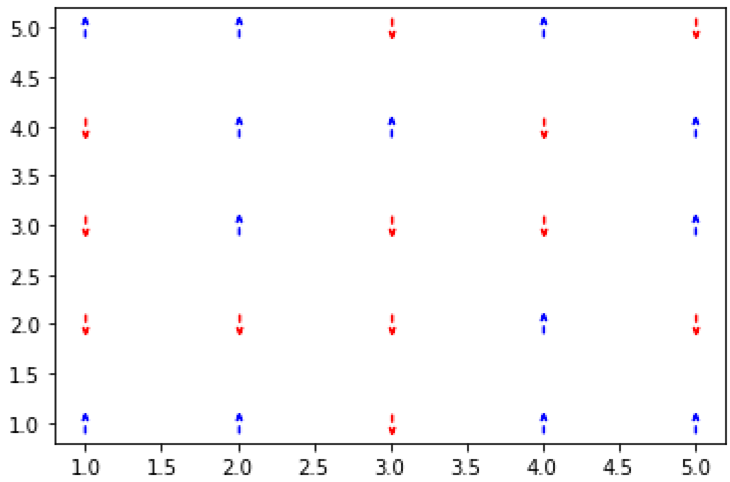


Figure 7: Order parameter S_i related to time-reversal symmetry for a lattice 5x5, taken at $T = 10K$ and for ten thousand simulations

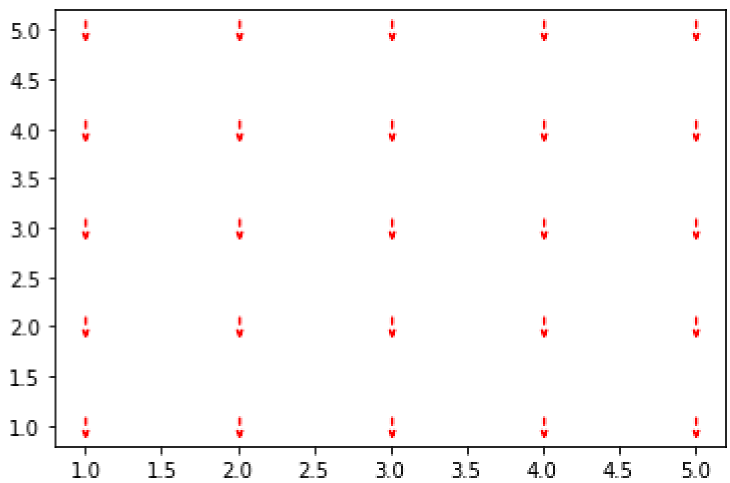


Figure 8: Order parameter S_i related to time-reversal symmetry for a lattice 5x5, taken at $T = 0.1K$ and for ten thousand simulations

The graph for $10K$ is showing a normal time-reversal symmetry state of the material, where, none of the sites are having the preferential configuration, meaning that the system is not showing any order parameter. The code is here returning $\langle|S|\rangle = 0.005$.

But, the second plot for $0.1K$ is showing a highly favored system completely ordered. Where all the phase orders are aligned and thus the system is showing an order parameter across the entire lattice. That is meaning that the system is in a time-reversal symmetry-breaking state (spontaneously). Also, the code is printing $\langle|S|\rangle = 1$. Note that the entire XY model could have been a blue down arrow.

As explained above, the results would have needed many more plots to get an accurate value of the critical temperature T_C. As well, we don't know from how many simulations the Monte Carlo method is showing good results. And how long does it take for this method to find a favored configuration for the system and hence an accurate average order parameter $\langle|S|\rangle$ for a given temperature.
Thus, the next section is a discussion on the liability and errors engendered by this statis-

tical method.

3.3 Flaws, error, and accuracy

By definition, statistics are however not giving exact results. Then, to end up with credible work, determining a range of flaws is necessary. This error scale is depending on two parameters: the number of simulations needed to approach an accurate value, and the spreading of a wrong-chosen value on all the samples.

The Monte Carlo approximation is conversely approaching the accurate value S for a given temperature T, over the total number of simulations N: $\langle |S| \rangle \propto \frac{1}{N}$.

However, the way S is getting closer to its accurate value is precisely given by:

$$\langle |S| \rangle = \frac{1}{N} \sum_{n=1}^{N} |S_n|$$

Where N is the number of simulation/measurement and S_n is the n-th measured value of S.

Hence, the standard deviation of S is describing the error depending on the number of simulations. Its expression is:

$$\sigma = \sqrt{\frac{\langle |S|^2 \rangle - \langle |S| \rangle^2}{N}}$$

Where,

$$\langle |S|^2 \rangle = \frac{1}{N} \sum_{n=1}^{N} |S_n|^2$$

This assumes that measurements are independent, which is probably not true. This will however be treated later.

The code used in 3.1 has been modified to output a graph showing this standard deviation over the total number of simulations returned by the Monte Carlo method. We hence plot the standard deviation for $N = 500$, $N = 2500$, and $N = 5000$ as an example.

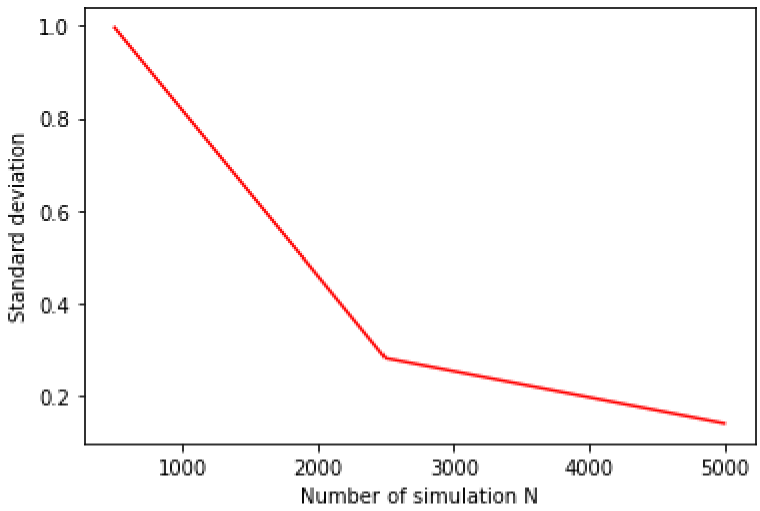


Figure 9: Standard deviation for $N = 500$, $N = 2500$ and $N = 5000$, taken at $T = 0.1K$

The magnitude of this standard deviation is getting smaller the more total simulation there is: which is making sense as this is the purpose and the result expected for a Monte Carlo approximation. This could however make a difference when calculating S for many different temperatures.

However, the standard deviation formula precisely that every value is independently generated from one other, which is not true. The S values are more and more correlated with each other the more there are measurements.

We can nevertheless quantify these correlations between the simulations by defining a function c:

$$c(l) = \sqrt{\frac{1}{N-l}\sum_{n=1}^{N-l}(S_n - \langle S\rangle)(S_{n+l} - \langle S\rangle)}$$

where $(|S_n| - \langle |S| \rangle)$ is a measure of the fluctuation of S about its average value. $c(l)$ is a function returning the correlation of these fluctuation for S values l measurement apart [19]. That is to say if S_n and S_{n+l} are independent, then $c(l) = 0$. Also, if there is no link between the values contained from n to l, the $c(l)$ fluctuation function is taking the form of the regular standard deviation, $c(l) = \sigma$.

You can as well note that, as in the example shown we took $\langle S\rangle = 0$ the further we go in the simulations. That is meaning that we would have the correlation function:

$$c(l) = \sqrt{\frac{1}{N-l}\sum_{n=1}^{N-l} S_n S_{n+l}}$$

And would highly simplify the expression of the error scale.

We use the code described above in 3.1 to plot this quantity depending on the value approached by the simulation $\langle |S| \rangle$. In the below plots, we took $l = 1$, meaning that $c(l)$ is expressing the correlation between the previous and the next value returned by the Monte Carlo simulation.

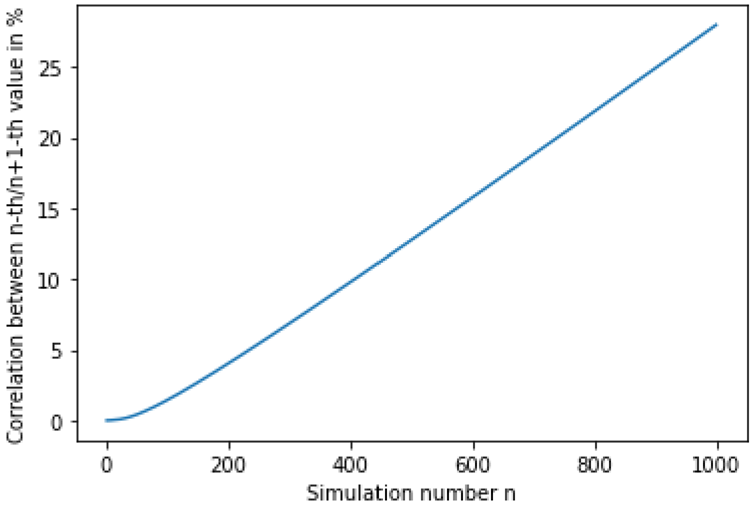


Figure 10: Amount of correlation between the n-th and the $n + 1$-th value in $\%$ for a thousand simulations

This plot for a thousand simulations is taking a linear form. And is demonstrating that the more simulation there is, the more the correlation between the values is high. This is making sense since after several steps we approach an accurate value, and the next step is comparing the previous value with a proposed one. In Figure 10, we should reach 100% for $N = 4000$ simulations. Thus, this graph is supposed to get flatter as it is reaching its equilibrium at $N \approx 4000$. Finally, the results acquired in 3.2 for 5000 simulations are hence credible.

Considering the standard deviation and the link between the values at each step, the spreading of a bad proposal and the steps needed to approach the accurate value has been characterized.

Such numerical work is having an error flow that is given by the statistical method used. Here we have defined the flaws generated by the Monte Carlo approach. Obviously, for a statistical technique like Monte Carlo, the more simulation there is, the more we are getting closer to an accurate value. However, this method is numerically limited as a high number of simulations means a lot of data to treat, which can easily become heavy work for a computer.

The minimum amount of simulation required to get meaningful results is difficult to determine as it depends on the results wanted. However, this work is trying to expose an explanation and an interpretation of those results.

4 Discussion: Interpretation of the results

4.1 Interpretation

The results acquired are leading to a physical explanation which is detailed in this section. The accuracy of the data generated explained in 3.3 by the program is making the interpretation credible. But please keep in mind that, the more there is a simulation sample, the more this numerical work, and hence this discussion would be representing an accurate physical behavior of the superconductive state.

On the results acquired in Figure 4, a separation between the spontaneously broken time-reversal symmetry and the normal time-reversal symmetry is visible as the order parameter $\langle |S| \rangle$ is contained between 1 and 0 included. When this order parameter is reaching 0, that means there is no anymore tendency for the system to create an order parameter across the component, then the spontaneously broken time-reversal symmetry has reached its critical temperature.

Below a critical temperature, both time-reversal symmetry and phase stiffness are affected (T_C need not be the same for both, however). Nonetheless, we have only simulated the order parameter S related to time-reversal symmetry breaking. It would have hence needed to calculate the phase stiffness to identify a superconductive state.

With the temperature increasing we can observe the order parameter S decreasing, showing a phase transition. However, we can't determine T_C, the separation between normal and spontaneously broken time-reversal symmetry state, from a single program run with a single lattice size. Also, T_c has to be taken as an example: as said before, the value kept on the graph is not accurate and would need many more program runs with systems of different lattices size.

Another comment on the result is that the graph is a bit jagged-looking: this can be associated with the fact that it would need many more steps with more simulations. Nevertheless, this was made difficult due to the time taken by the program to have the simulations done, as the program was written in Python.

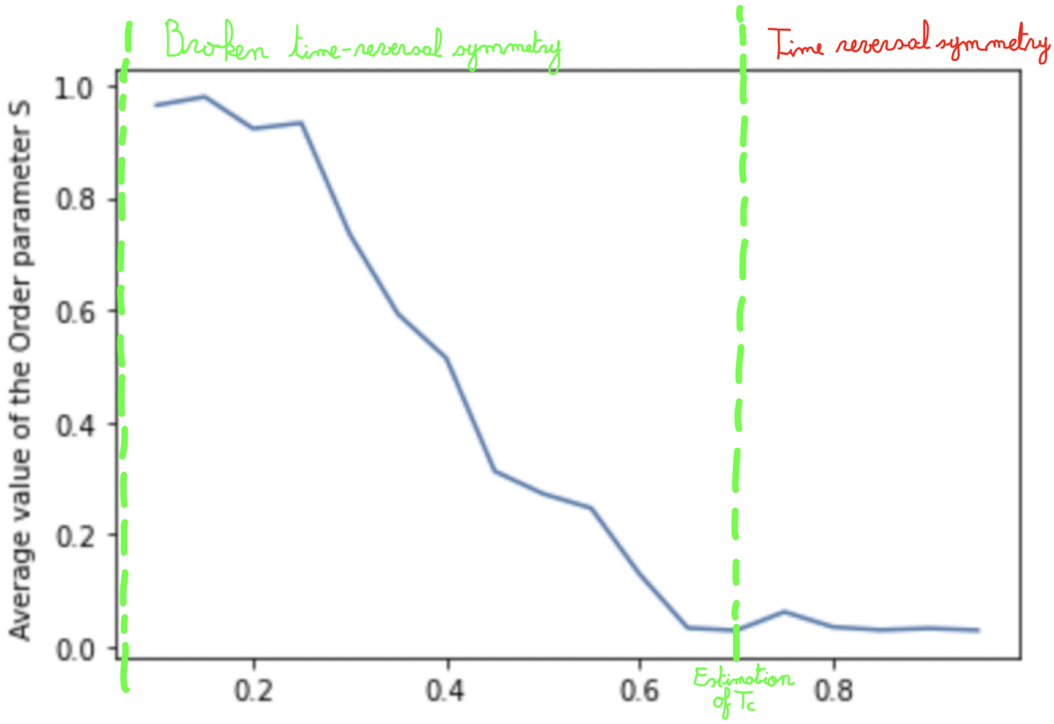


Figure 11: Separation of the normal and superconductive state on figure 4

You can note as well that there is still a little bit of noise in the time-reversal symmetry section because the system is in a finite size.

As Figure 6 is showing a lattice fully with the same order of the phases, we can deduce this alignment of the phase is due to the behavior of the electron: Cooper pairs are forming a zero total spin: a spin of one electron is up and the other is down, due to the Pauli exclusion principle. We have seen in 1.3.1, that this alignment is affecting to broken time-reversal symmetry. Now if all spins of electrons are paired in up and down direction, that is meaning that we can't reverse in time this tendency: if we apply the time-reversal operator, all the spin will point in the opposite direction from the initial direction. This is resulting from $TST^{-1} = -S$ as S (the spin) is complex (S_y is complex). That will cause in the case of the GL-theory a non-zero order parameter $\langle |S| \rangle \neq 0$ satisfying $TS = -S$ (S is the order parameter), characterizing the spontaneously broken time-reversal symmetry. In a normal state, as all the spin doesn't have any preferential pointing, a state given can be reversed in time.

As it would have needed to simulate numerically the order parameter related to the phase stiffness to explain a superconductive state, we can't end up interpreting the results by having characterized superconductivity. However, the method to calculate the order parameter for time-reversal symmetry has been shown. This spontaneously broken symmetry is then engendered by the alignment of the phase order described by the Boltzmann factor.

Nonetheless, the time-reversal symmetry and phase stiffness are very important characteristics to observe whether a material is in a superconductive state or not. Determining the precise temperature T_C where time-reversal symmetry and phase stiffness are spontaneously broken is important to classify 2D multi-components depending on their amplitude for inter-band coupling (η).

This critical temperature can rise interesting applications on superconductivity as it is depending on the ease to keep the material below this critical value and then maintain the component in a superconductive state. Some different applications and fields of research are presented in the next section to link this project to other researchers' work.

4.2 Conclusion

Now give an end to this work, after displaying the results and how it should be acknowledged. This section is a summary of what should be reminded of this work and what could have been better. An opening to other works from different authors, or other research on superconductivity that could be looked at, is also presented.

Even though we have only calculated the order parameter related to time-reversal symmetry, several notions about superconductivity can be remembered. Thus, what is to be reminded from this work is that the 2D superconductive state is characterized by both breaking time-reversal symmetries and phase stiffness, engendered by Cooper pairs forming an order parameter with more or less coherence length. Also, all of this is happening under a critical temperature T_C that can be determined numerically.

The results acquired numerically in this work are showing the method to determine the critical temperature through simulation while approaching the physical behavior of a superconductive system. However, to fully exploit superconductors for engineering purposes, this value is very important to determine which material is easier to keep in its superconductive state. Several applications of superconductors are highly studied notably using the Meissner effect characterized by the penetration depth (or the ratio between the coherence length ϵ and the penetration depth λ, defined as $\kappa = \frac{\lambda}{\epsilon}$) as it is the penetration possible by a magnetic field until the perturbation of the superconductor.

Superconductors are already used in many fields: medical MRI/NMR devices, magnetic-energy storage systems, motors, generators, transformers, computer parts, and sensitive devices for the measurement of magnetic fields, and electrical currents. Those are devices using the magnetic field generated continuously by superconductive components. As well as futuristic devices are the main concern of research groups, for instance, electric power transmission, electric motors, magnetic levitation devices, and superconducting magnetic refrigerators.

Helium super-fluid is also a major discovery related to superconductivity. For this, helium liquid (helium at $4K$) is happening to cross ceramic as it is showing a single macroscopic wave function that is not affected by friction and can slide between the holes of the ceramic. That is also meaning that if we rotate the liquid at a certain speed, the liquid can rotate indefinitely as it is not showing any effects from friction.

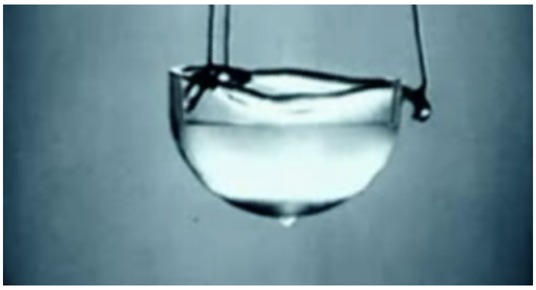


Figure 12: Picture of superfluid liquid helium, taken from [20]

Finally, for further studies on the breaking of time-reversal and $U(1)$ symmetry in 3D multi-components, you can refer to both articles:

-"Time-Reversal Symmetry Breaking and Spontaneous Anomalous Hall Effect in Fermi Fluids" by Kai Sun and Eduardo Fradkin published in Physics Review B and Research Gate.
-Spontaneous breakdown of time-reversal symmetry induced by thermal fluctuations by Johan Carlström and Egor Babaev published on Physics review B.

The first article is an application of this work on the Hall effect and the second is a more precise study of time-reversal and $U(1)$ symmetry breaking with vortices created by a gauge field.

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